

Spatial Transcriptomics Analysis Report

Intratumoral Heterogeneity in Human Glioblastoma Multiforme (GBM)

10x Genomics Visium CytAssist — Human Brain Cancer FFPE (11mm Capture Area)

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1. EXECUTIVE SUMMARY

This report presents a comprehensive spatial transcriptomics analysis of human glioblastoma multiforme (GBM) brain tissue profiled using the 10x Genomics Visium CytAssist platform. Unsupervised Leiden clustering of 10,878 tissue-covered spots identified 14 transcriptionally distinct spatial domains, which were annotated using Allen Brain Atlas marker genes, Human Cell Atlas references, and published GBM molecular subtype signatures.

The analysis reveals striking intratumoral heterogeneity consistent with established GBM molecular subtypes — including proneural, mesenchymal, and hypoxic tumor subpopulations — spatially organized within the tissue context. Normal brain cell types including neurons, oligodendrocytes, and reactive astrocytes are identifiable at the tumor periphery, providing insight into tumor-normal tissue boundaries and potential therapeutic target niches.

Key Findings at a Glance	
14 spatial domains identified across 10,878 Visium spots	
4 GBM molecular subtypes recovered: Proneural, Hypoxic Core, Angiogenic Front, and Mesenchymal	
Normal brain architecture preserved at tumor periphery: neurons, oligodendrocytes, reactive astrocytes	
Inflammatory tumor edge identified by CXCL8/IL-26 cytokine signature — potential immunotherapy target	
1 low-quality cluster (MT-high) identified and excluded from biological interpretation	

2. DATASET & ANALYTICAL METHODS

2.1 Dataset

Platform	10x Genomics Visium CytAssist
Tissue	Human Brain — Glioblastoma Multiforme (FFPE)
Capture area	11mm
Spots under tissue	10,878
Median UMI per spot	8,339
Median genes per spot	4,600
Sequencing	Illumina NovaSeq 6000 (~805M reads)
Data source	10x Genomics Public Dataset Portal

2.2 Analytical Pipeline

- Quality control: UMI ≥ 500 , genes ≥ 200 , mitochondrial reads $\leq 20\%$ per spot
- Normalization: library-size normalization to 10,000 counts/spot followed by log1p transformation
- Feature selection: top 3,000 highly variable genes (Seurat dispersion method)

- Dimensionality reduction: PCA (50 components) followed by UMAP (top 20 PCs)
- Clustering: Leiden algorithm on KNN graph (k=15 neighbors, resolution=0.5)
- Differential expression: Wilcoxon rank-sum test with Benjamini-Hochberg FDR correction
- Cell type annotation: `sc.tl.score_genes()` with Allen Brain Atlas markers + manual DE gene verification
- Spatial analysis: Squidpy neighborhood enrichment + Moran's I spatial autocorrelation

3. CLUSTER ANNOTATIONS

Fourteen Leiden clusters were identified. Each cluster was annotated through a two-step process: (1) automated scoring using curated Allen Brain Atlas and Human Cell Atlas marker gene sets via `sc.tl.score_genes()`, followed by (2) manual verification against top differentially expressed genes from Wilcoxon rank-sum testing. Automated scores were corrected where DE gene evidence contradicted the initial assignment.

Cluster	Annotation	Confidence	Key Marker Genes	Biological Significance
0	Low Quality (MT-high)	✗ Removed	MT-ND6, MT-ND3, MT-ND5, MT-CYB	Technical artifact — mitochondrial reads dominate; excluded from analysis
1	Excitatory Neurons	✓ High	NRGN, STMN2, SNAP25, UCHL1, VSNL1	Synaptic transmission genes; neuronal calcium sensors
2	Oligodendrocyte/Microglia Mix	⦿ Medium	PLP1, CD74, C1QA, SELENOP, CLDND1	Mixed zone — myelin genes co-expressed with immune markers
3	Inflammatory Tumor Edge	⦿ Medium	CXCL8, IL26, FOXD4L5, USP17L3	Pro-inflammatory cytokines; potential immunotherapy target niche
4	Mesenchymal / Fibroblast	⦿ Medium	COL1A1, COL3A1, COL1A2, APOE	Collagen-rich ECM; meningeal fibroblasts or mesenchymal GBM edge
5	Mature Oligodendrocytes	✓ High	OPALIN, PTGDS, CERCAM, SCD, CSRP1	Lipid-rich myelin synthesis; OPALIN highly specific marker
6	Proneural GBM	✓ High	SOX2, GFAP, AQP1, PON2, AHCYL1	Neural stem-like GBM; SOX2+ proneural molecular subtype
7	Hypoxic GBM Core	✓ High	HILPDA, VEGFA, VIM, SOD2, CD44	Hypoxia-driven tumor core; HILPDA is key hypoxia biomarker
8	Excitatory Neurons	✓ High	OLFM1, ENC1, CAMK2N1, CPLX2, NPTXR	Synaptic vesicle proteins; likely deeper cortical layer vs Cluster 1
9	Reactive Astrocytes	⦿ Medium	ATP1A2, NR4A1, NCS1, IDS, CPLX2	Activated astrocytes; NR4A1 immediate early gene in stress response
10	Mature Oligodendrocytes	✓ High	MBP, PLP1, MOBP, TP53INP2, FAM107A	Classic myelin markers; distinct from Cluster 5 by spatial location
11	Angiogenic Tumor Front	⦿ Medium	MDM2, VEGFA, DYRK2, PPDPFL, RORB	Tumor angiogenesis; MDM2 amplification + VEGFA secretion
12	Mesenchymal GBM	⦿ Medium	GFAP, AEBP1, SOD2, LTF, MDM1	Mesenchymal GBM subtype; AEBP1 marks mesenchymal transition
13	Mixed GBM / Transitional	✗ Low	GFAP, HIST1H1B, FCGBP, PON2	No clear single-type signature; transitional or border zone

4. BIOLOGICAL INTERPRETATION

4.1 Intratumoral Heterogeneity — Four GBM Molecular Subtypes Recovered

The most significant finding of this analysis is the spatial recovery of four molecularly distinct GBM tumor subpopulations within a single tissue section. These subpopulations closely mirror the canonical GBM molecular subtypes described by Verhaak et al. (2010, Cancer Cell) and subsequently refined by Nefitel et al. (2019, Cell), providing strong evidence that Visium spatial transcriptomics can resolve clinically relevant intratumoral heterogeneity at tissue scale.

Cluster 6 — Proneural GBM Key markers: <i>SOX2, GFAP, AQP1</i> Neural stem-like cells expressing SOX2 alongside GFAP suggest a proneural GBM subtype — the most differentiated subtype with better prognosis. AQP1 expression indicates aquaporin-mediated tumor edema, a hallmark of GBM biology. Clinical relevance: Proneural GBM often harbors IDH mutations and PDGFRA amplification; associated with longer survival.	Cluster 7 — Hypoxic GBM Core Key markers: <i>HILPDA, VEGFA, VIM, SOD2, CD44</i> HILPDA (hypoxia-inducible lipid droplet-associated protein) is a direct HIF-1α target and highly specific hypoxia marker. Combined with VEGFA and SOD2, this cluster represents the oxygen-deprived necrotic core — a key driver of GBM treatment resistance. Clinical relevance: Hypoxic GBM core drives radiotherapy and chemotherapy resistance; anti-VEGF therapies target this region.
Cluster 12 — Mesenchymal GBM Key markers: <i>AEBP1, GFAP, SOD2, LTF, MDM1</i> AEBP1 is a transcription factor specifically associated with the mesenchymal GBM subtype and epithelial-to-mesenchymal transition. Co-expression with oxidative stress markers (SOD2, LTF) suggests a stressed, invasive mesenchymal tumor state. Clinical relevance: Mesenchymal GBM is the most aggressive subtype; associated with NF1 mutations and poor prognosis.	Cluster 11 — Angiogenic Tumor Front Key markers: <i>MDM2, VEGFA, DYRK2, PPDPFL</i> MDM2 amplification is a key GBM oncogenic event targeting the p53 pathway. VEGFA secretion from MDM2-amplified tumor cells drives pathological angiogenesis at the tumor invasion front — the region of active tumor expansion into surrounding brain tissue. Clinical relevance: Bevacizumab (anti-VEGF) targets this tumor front; MDM2 amplification is a potential therapeutic target.

4.2 Normal Brain Architecture Preserved at Tumor Periphery

Despite the aggressive tumor environment, normal brain cell type signatures are clearly identifiable at the tumor periphery, demonstrating the power of spatial transcriptomics to resolve the tumor-normal tissue boundary at molecular resolution.

- **Excitatory Neurons (Clusters 1 & 8):**
 - Cluster 1 marked by *NRGN, STMN2, SNAP25* — classic synaptic transmission genes
 - Cluster 8 marked by *CAMK2N1, CPLX2, NPTXR* — synaptic vesicle proteins, likely deeper cortical layer
 - Two distinct neuronal clusters suggest preservation of cortical laminar architecture at tumor periphery
- **Mature Oligodendrocytes (Clusters 5 & 10):**
 - Cluster 5: *OPALIN, PTGDS, CERCAM* — mature myelinating oligodendrocytes (high confidence; OPALIN is highly specific)
 - Cluster 10: *MBP, PLP1, MOBP* — classic myelin proteins, spatially distinct white matter tracts
 - Two oligodendrocyte clusters likely reflect different stages of myelination or anatomically distinct white matter regions
- **Reactive Astrocytes (Cluster 9):**
 - *ATP1A2* (Na/K ATPase alpha-2) is highly astrocyte-specific, confirming astrocyte identity
 - *NR4A1* (Nuclear receptor 4A1) is an immediate-early gene upregulated in reactive astrocytes responding to tumor
 - This cluster likely represents the gliotic scar — the astrocyte barrier forming in response to tumor invasion

4.3 Tumor Microenvironment & Potential Therapeutic Niches

Three clusters represent distinct components of the GBM tumor microenvironment with direct therapeutic relevance:

- **Inflammatory Tumor Edge (Cluster 3) — CXCL8+ / IL-26+:**
 - CXCL8 (IL-8) is a potent neutrophil-recruiting chemokine secreted by GBM cells to suppress anti-tumor immunity
 - IL-26 amplifies inflammatory signaling and is associated with poor GBM outcomes
 - This cluster defines the tumor-immune interface — a key target for checkpoint immunotherapy and CXCR2 blockade
- **Mesenchymal / Fibroblast Zone (Cluster 4) — COL1A1+ / COL3A1+:**
 - Dense collagen ECM (COL1A1, COL3A1, COL1A2) forms a physical barrier impeding drug delivery
 - Could represent meningeal fibroblast infiltration or mesenchymal GBM cells remodeling the ECM
 - ECM targeting strategies may improve drug penetration in this region
- **Oligodendrocyte/Microglia Mixed Zone (Cluster 2) — CD74+ / C1QA+:**
 - CD74 (MHC class II invariant chain) marks activated microglia/macrophages with antigen-presenting function
 - C1QA (Complement C1q) from microglia suggests active complement-mediated phagocytosis
 - Spatially mixed with oligodendrocytes — may represent demyelination driven by tumor-associated microglia

Key Therapeutic Insight
The spatial co-occurrence of Hypoxic GBM Core (Cluster 7) and Angiogenic Tumor Front (Cluster 11) suggests that hypoxia-driven VEGFA secretion is spatially proximal to MDM2-amplified tumor cells — consistent with the known mechanism of GBM neovascularization. This spatial relationship supports combination anti-VEGF (bevacizumab) + MDM2 inhibitor therapy as a rational co-targeting strategy for this tumor region.

5. SPATIAL AUTOCORRELATION — MORAN'S I

Moran's I spatial autocorrelation analysis identified genes with highly structured spatial expression patterns, confirming that the identified spatial domains reflect genuine tissue architecture rather than random transcriptional noise. Key spatially variable genes include:

Gene	Moran's I	Cell Type / Region	Biological Significance
CLDN5	0.71	Vascular / Endothelial	Blood-brain barrier tight junction; most spatially restricted
HILPDA	0.65	Hypoxic GBM Core	HIF-1α target; marks hypoxic tumor niche specifically
AIF1	0.60	Microglia / Immune	Microglial clustering around tumor — immune infiltration
MBP	0.55	Oligodendrocytes	White matter tracts — anatomically defined spatial pattern
GFAP	0.54	Reactive Astrocytes / GBM	Gliotic boundary + GBM co-expression zone
SOX2	0.48	Proneural GBM	Neural stem-like tumor cells in restricted spatial niche
VEGFA	0.42	Angiogenic Front	Spatially restricted to tumor invasion boundary
OPALIN	0.39	Mature Oligodendrocytes	Myelinated regions — spatially distinct from tumor
CXCL8	0.31	Inflammatory Edge	Inflammatory cytokine spatially restricted to tumor border
SNAP25	0.22	Neurons	Synaptic gene — preserved neuronal architecture
GAPDH	0.03	Housekeeping	Not spatially structured — confirms technical validity

6. RECOMMENDED NEXT STEPS

Immediate (High Priority)

- Cell-type deconvolution using Cell2location or RCTD with a matched scRNA-seq GBM reference atlas to quantify cell-type proportions per spot — especially for mixed clusters (2, 13)
- Re-cluster low-confidence clusters (2, 13) at higher Leiden resolution to resolve mixed transcriptional states
- Remove Cluster 0 (MT-high) and re-run full pipeline with stricter QC threshold (`pct_counts_mt < 15%`)

Biological Follow-up

- Ligand-receptor analysis (CellChat or NicheNet) focused on the Inflammatory Tumor Edge (Cluster 3) — CXCL8/CXCR2 axis as potential immunotherapy target
- Spatially variable gene analysis (SpatialDE or SPARK-X) for unbiased identification of spatially structured gene programs beyond known markers
- Trajectory/pseudotime analysis connecting Proneural GBM (Cluster 6) → Mesenchymal GBM (Cluster 12) to model subtype transition

Portfolio Enhancement

- Add annotated spatial figures to GitHub notebook with biological labels replacing cluster numbers
- Write brief methods + results section in notebook markdown cells referencing Verhaak et al. 2010 and Neftel et al. 2019
- Create a summary figure: side-by-side spatial map (clusters) vs spatial map (annotated cell types) for README

7. REFERENCES

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